

Comparative Bulk RNA Sequencing Analysis of Brain, Muscle and Kidney Tissue Samples

Lorenzo Marceca

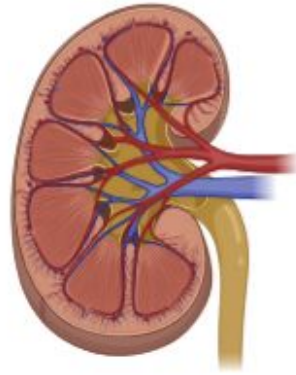
Tissues



Brain



Muscle



Kidney

Quality Control

- $RIN \geq 6$
- % of uniquely mapped $\geq 80\%$
- Fraction of rRNA reads $< 10\%$ (0.1)

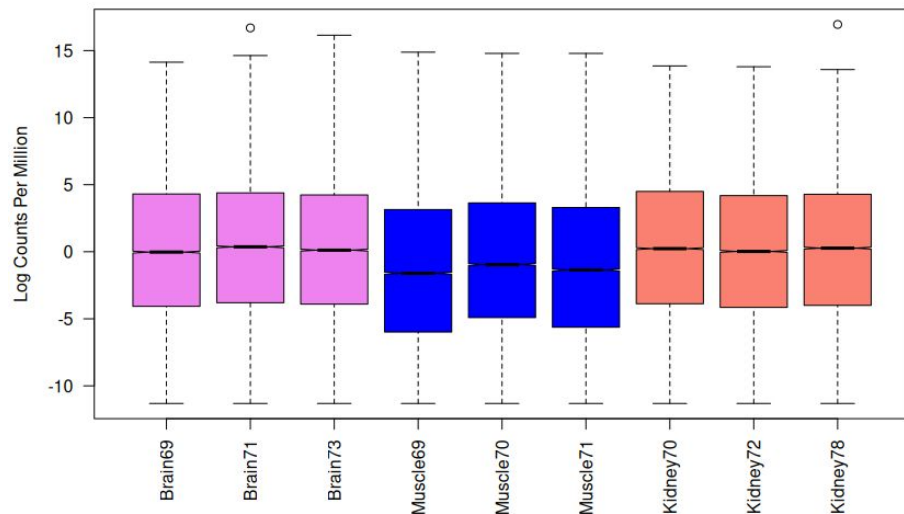
Sample	RIN	% mapped reads	fraction of rRNA reads
Brain69	7.5	89.9	0.015
Brain70	5.9	91.4	0.081
Brain71	6.2	91.0	0.079
Brain72	5.9	90.3	0.110
Brain73	6.8	88.4	0.067

Sample	RIN	% mapped reads	fraction of rRNA reads
Muscle69	8.6	90.8	0.017
Muscle70	6.9	88.1	0.02
Muscle71	8.4	89.0	0.012

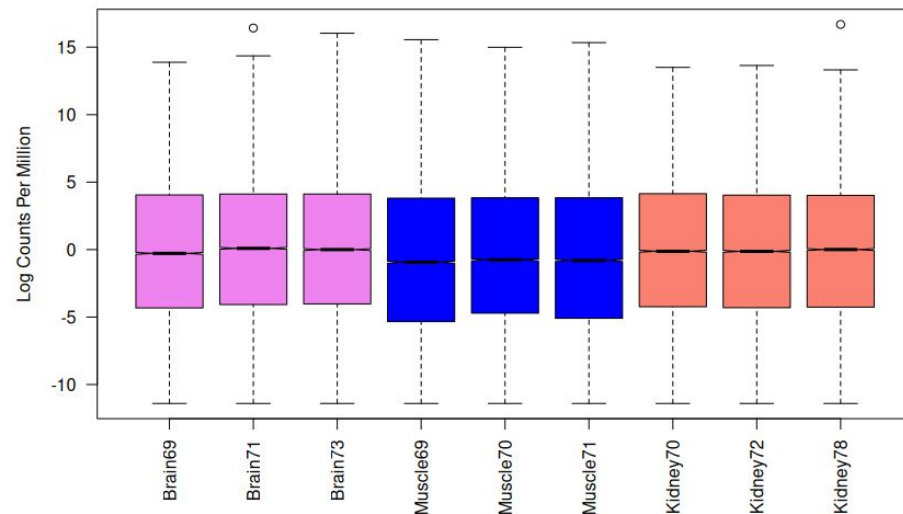
Sample	RIN	% mapped reads	fraction of rRNA reads
Kidney69	5.7	92.0	0.035
Kidney70	8.4	92.1	0.014
Kidney71	5.5	91.3	0.081
Kidney72	6.1	91.6	0.007
Kidney73	5.5	91.1	0.051
Kidney74	5.5	91.6	0.060
Kidney75	5.6	92.4	0.052
Kidney76	5.9	90.8	0.116
Kidney77	6.0	91.7	0.135
Kidney78	6.0	87.6	0.073

Normalization

Log-CPM Before TMM Normalization

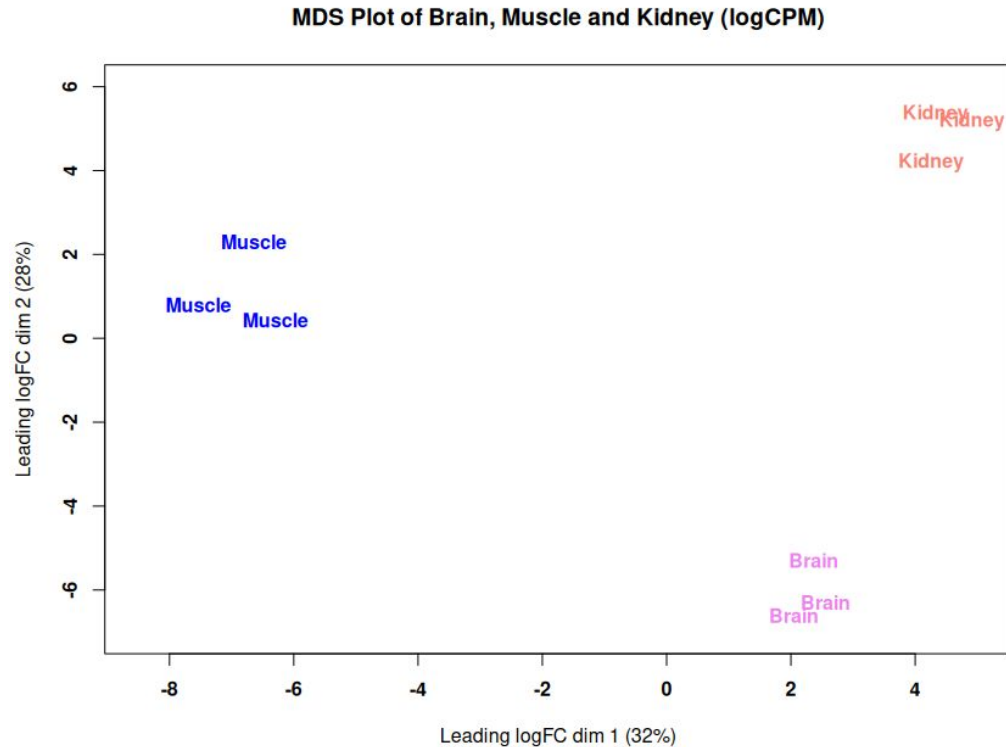


Log-CPM After TMM Normalization



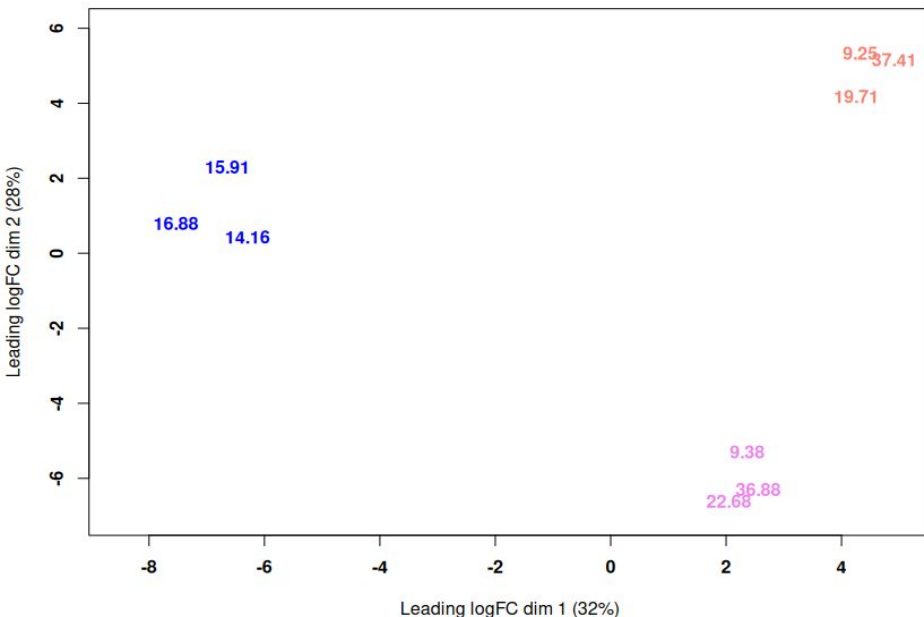
Brain69	Brain71	Brain73	Muscle69	Muscle70	Muscle71	Kidney70	Kidney72	Kidney78
1.1933	1.2072	1.0855	0.6282	0.8695	0.6840	1.2776	1.1134	1.2034

Multidimensional Scaling

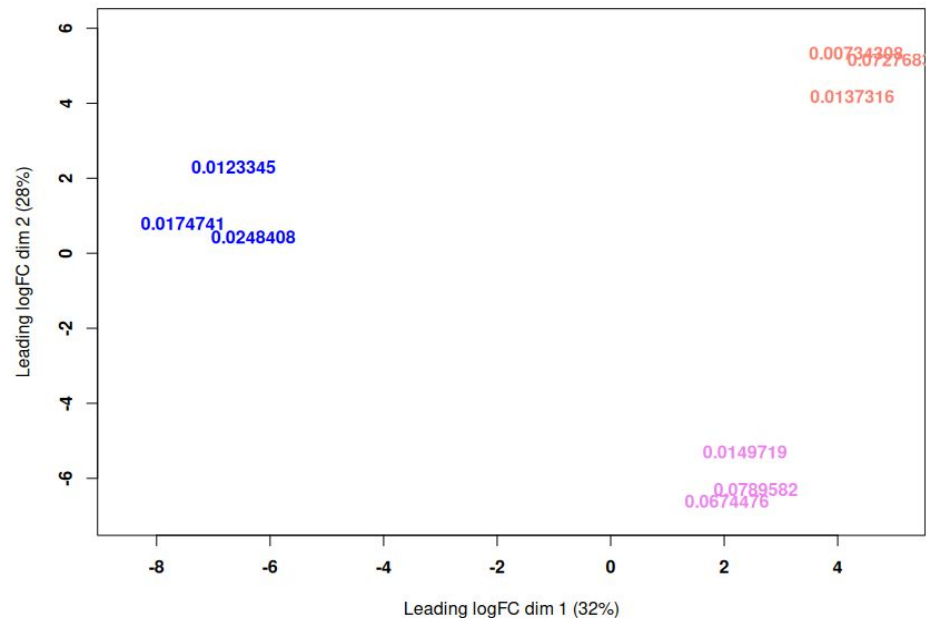


Comparing estimated % of mtRNA and rRNA reads

MDS Plot of Sample Groups (by estimated % of mtRNA reads)



MDS Plot of Sample Groups (by estimated fraction of rRNA reads)



Differentially Expressed Genes

Thresholds: FDR < 0.01 and |LogFC| > 1

Kidney vs Brain

Up	Down
1971	2180

Muscle vs Brain

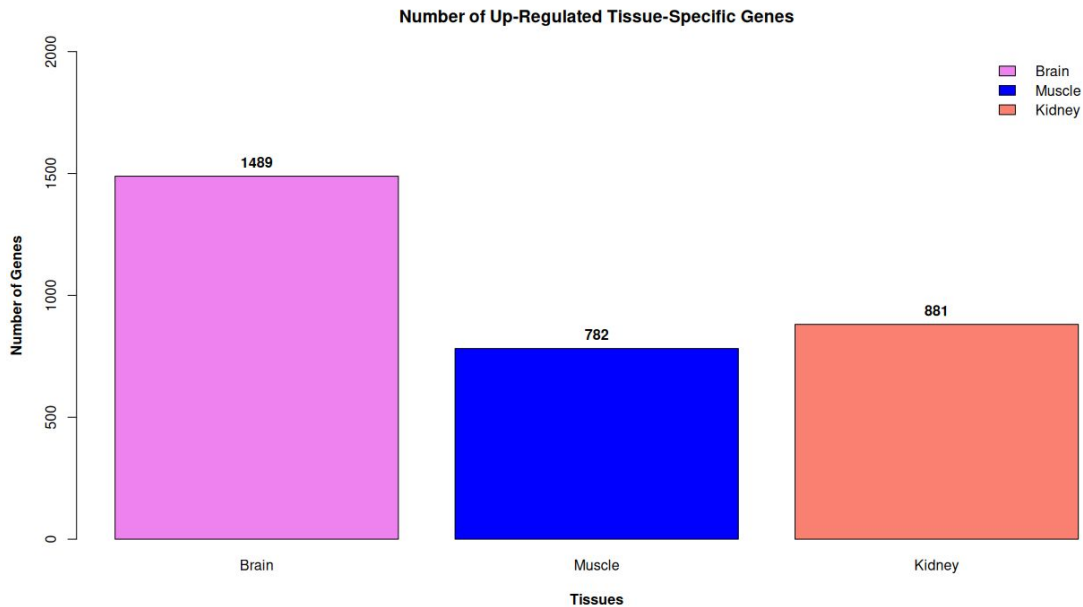
Up	Down
1975	3126

Muscle vs Kidney

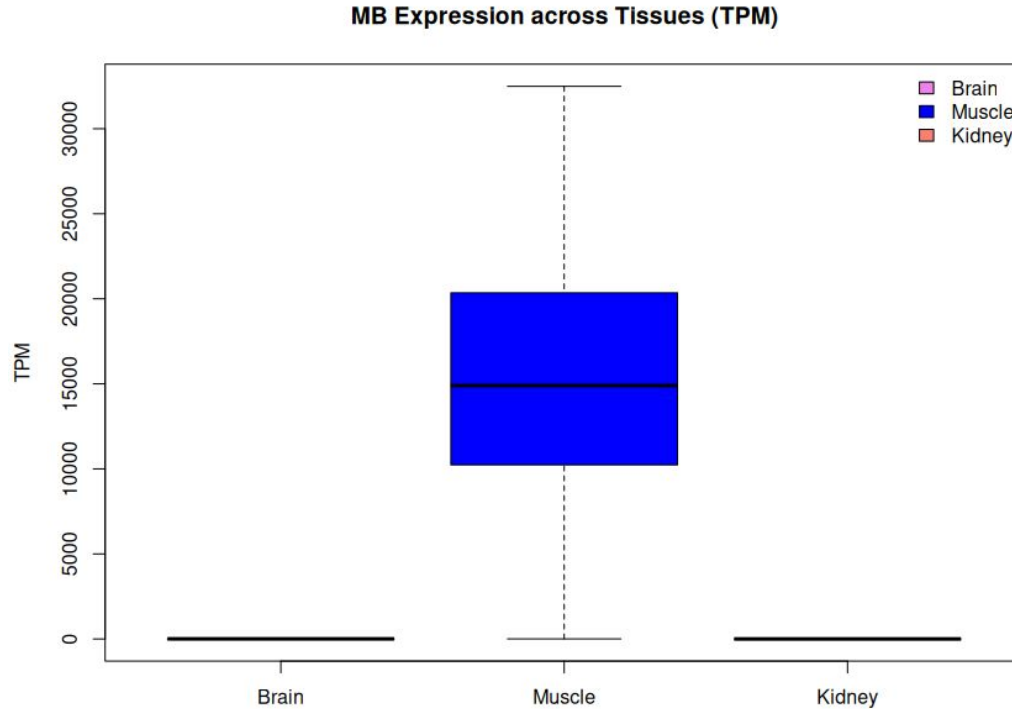
Up	Down
1296	2043

Differentially Expressed Genes

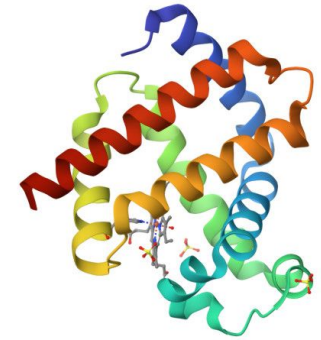
Thresholds: $FDR < 0.01$ and $|\text{LogFC}| > 1$



Expression of MB gene



Wilcoxon test (pairwise) to assess significance:
p-value < 2.2e-16



Crystal Structure of Myoglobin with Charge Neutralized Heme in *Homo sapiens*. Source RCSB-PDB

Functional enrichment analysis on DE genes



Tissue	GO Term	Description	Category	Adjusted p-value
Brain	GO:0043005	Neuron Projection	Cellular Component	2.750e-48
	GO:0007268	Chemical Synaptic Transmission	Biological Process	1.050e-47
	GO:0099634	Postsynaptic Specialization Membrane	Cellular Component	7.545e-43
Muscle	GO:0030239	Myofibril Assembly	Biological Process	1.167e-33
	GO:0031032	Actomyosin Structure Organization	Biological Process	4.151e-27
	GO:0045214	Sarcomere Organization	Biological Process	2.782e-26
Kidney	GO:0070062	Extracellular Exosome	Cellular Component	3.599e-48
	GO:1903561	Extracellular Vescicle	Cellular Component	1.196e-47
	GO:0008202	Steroid Metabolic Process	Biological Process	1.488e-9

Single-cell RNA Sequencing Analysis of *Homo Sapiens* Peripheral blood mononuclear cells

Lorenzo Marceca

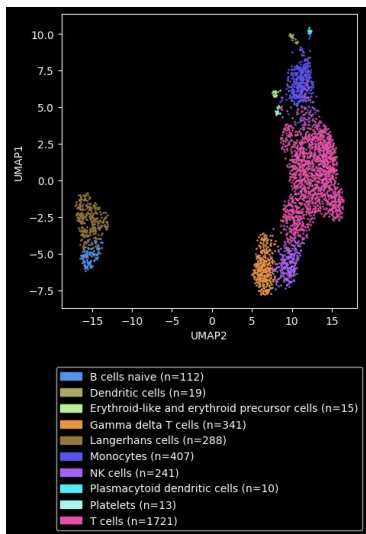
Dataset Overview

Comparison of clustering tools in R for medium-sized 10x Genomics single-cell RNA-sequencing data

Saskia Freytag^{1,2}, Luyi Tian^{2,3}, Ingrid Lönnstedt⁴, Milica Ng⁴, Melanie Bahlo^{1,2}

Affiliations + expand

PMID: 30228881 PMCID: PMC6124389 DOI: 10.12688/f1000research.15809.2



SRA: SRA713577
Protocol: 10x chromium
Cell number: 3334
Cluster number: 12

FILTERING
min features = 200
min cells = 3

3568 cells

Cell type reported	Putative Marker Gene
Naïve CD4+ T cells	IL7R, CCR7
Memory CD4+ T	CD3D, CD3E, IL7R
CD8+ T Cells	CD8A, CD8B
NK Cells	NKG7, GNLY, KLRF1
CD14+ Monocytes	CD14, LYZ, CST3
Naïve B Cells	MS4A1
Platelets	PPBP
Granulocytes / Neutrophils	FCGR3B, CXCR2, CXCL1
Plasmacytoid Dendritic Cells	LILRA4, CLEC4C
Erythroid-like Cells	HBB

Publication reference: <https://pubmed.ncbi.nlm.nih.gov/30228881/>

Quality control

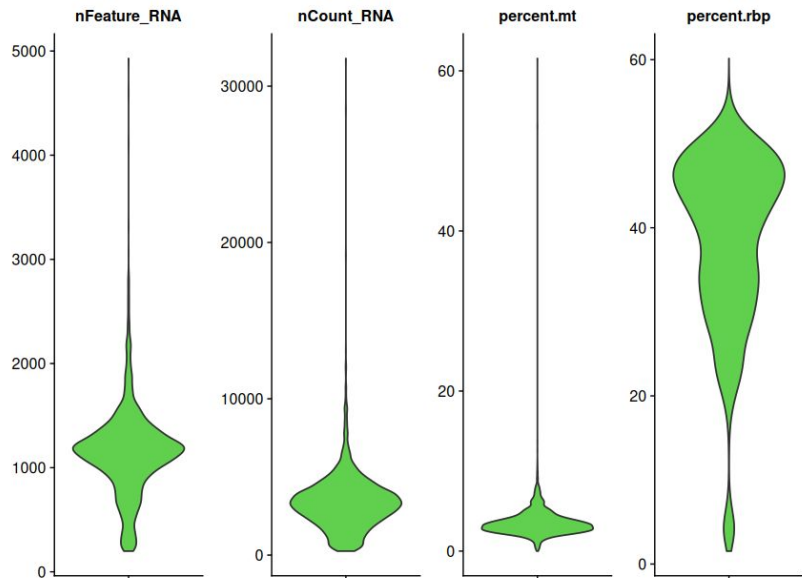
Thresholds:

- 200 < Number of genes < 2500
- Mitochondrial RNA < 5%

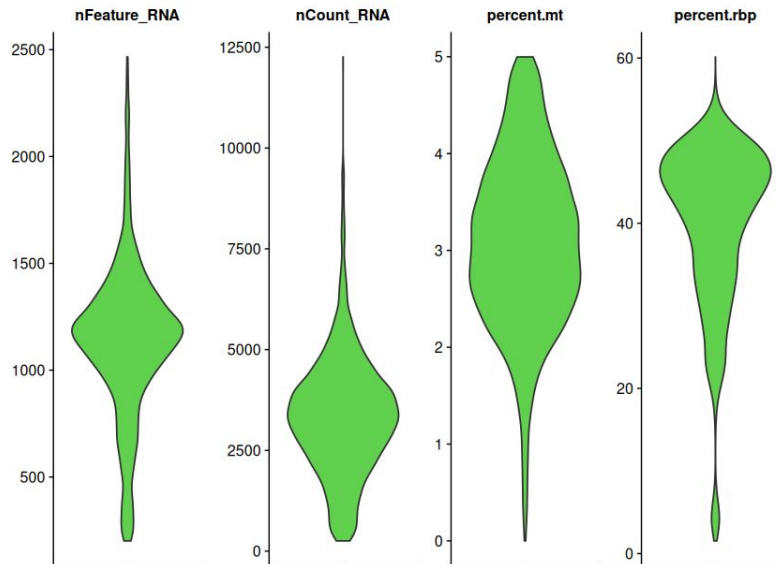


**86.7% (3095 / 3568) passed
quality assessment**

Quality metrics before Filtering

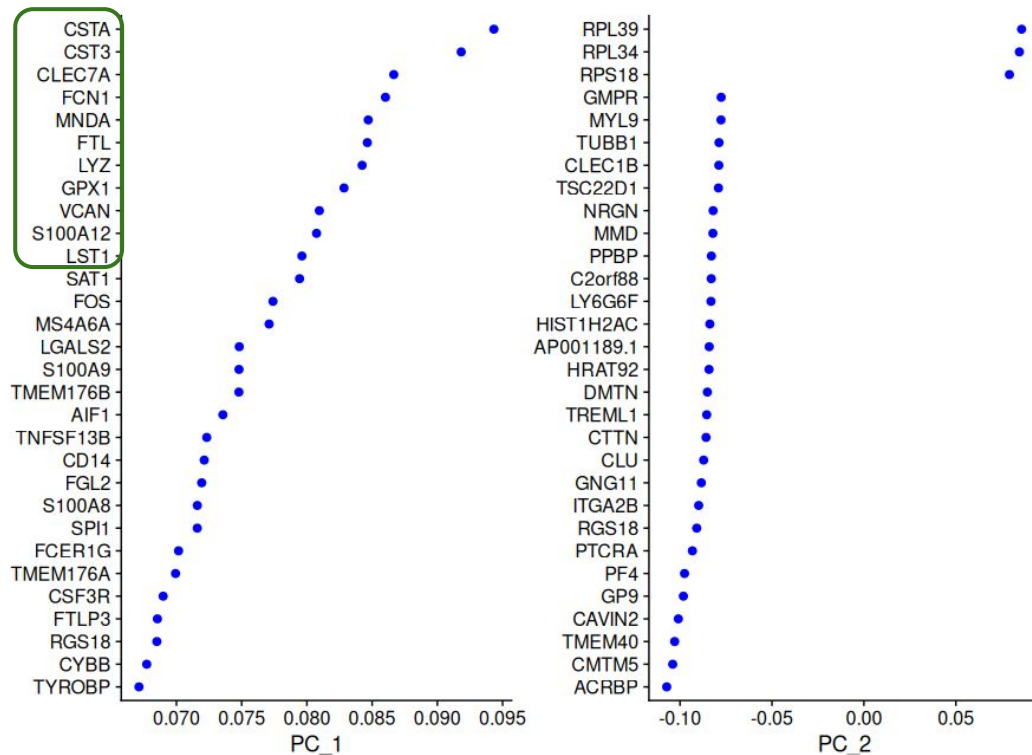


Quality metrics after Filtering

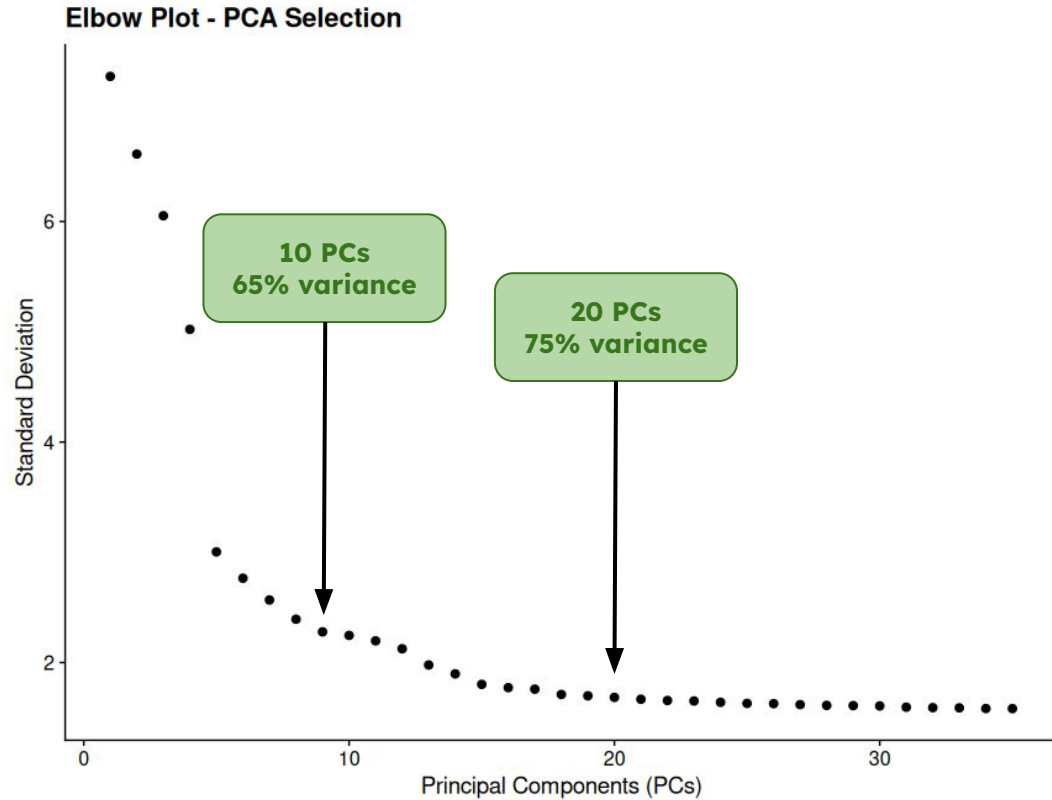


PCA

Monocytes

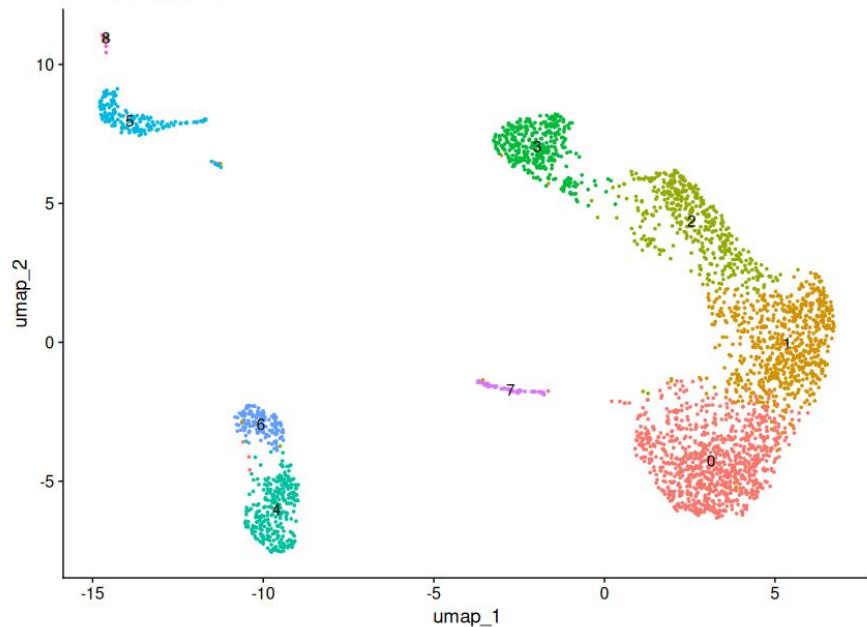


Elbow plot



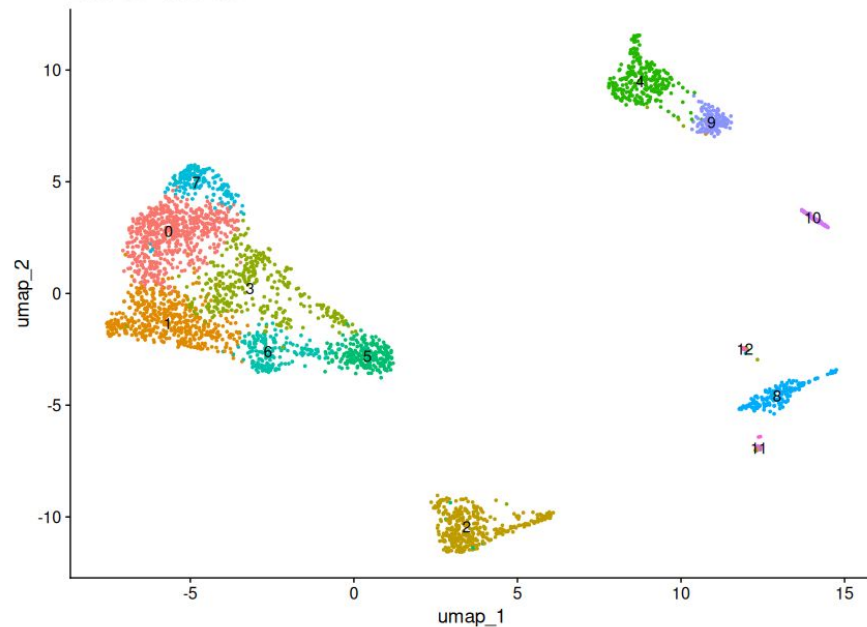
Clustering Results

10 PCs - Res 0.5



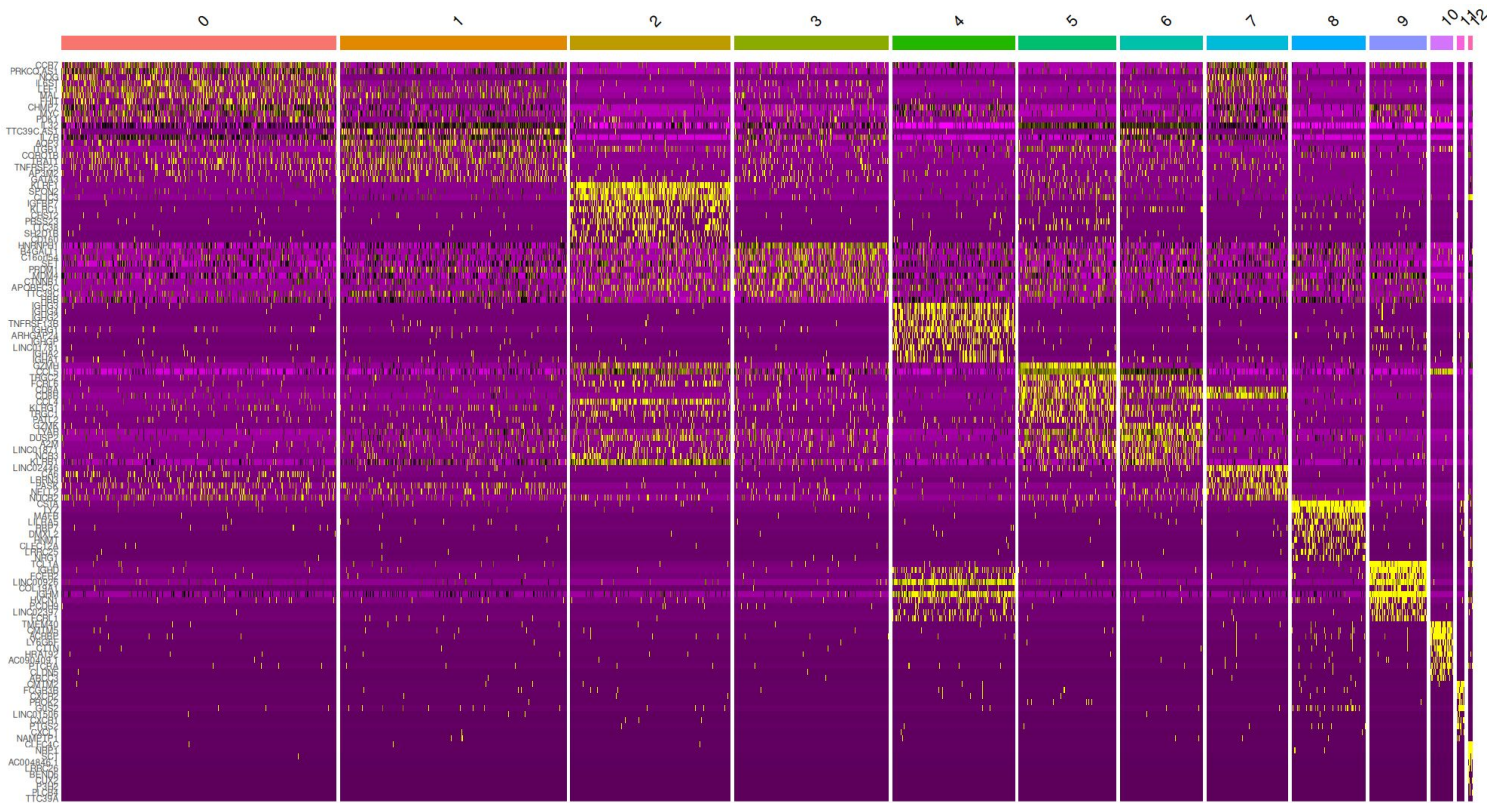
0	1	2	3	4	5	6	7	8
937	698	439	366	273	179	133	51	19

20 PCs - Res 0.8

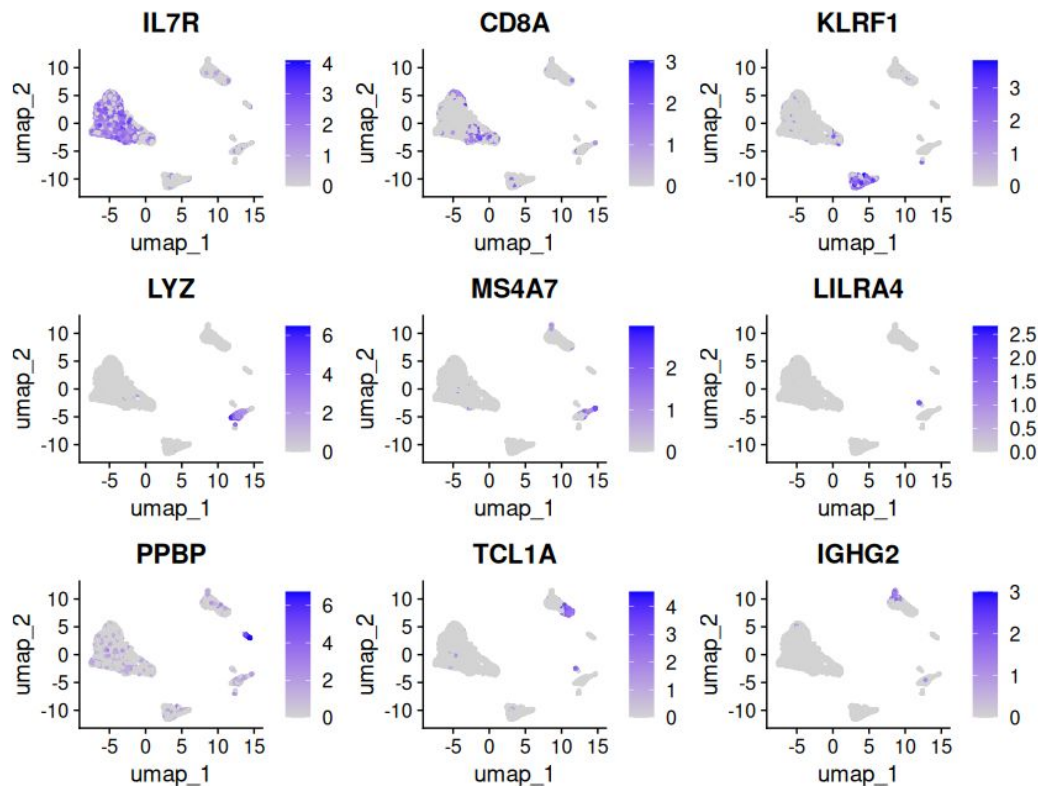


0	1	2	3	4	5	6	7	8	9	10	11	12
623	512	363	349	278	221	188	184	168	130	51	18	10

Heatmap

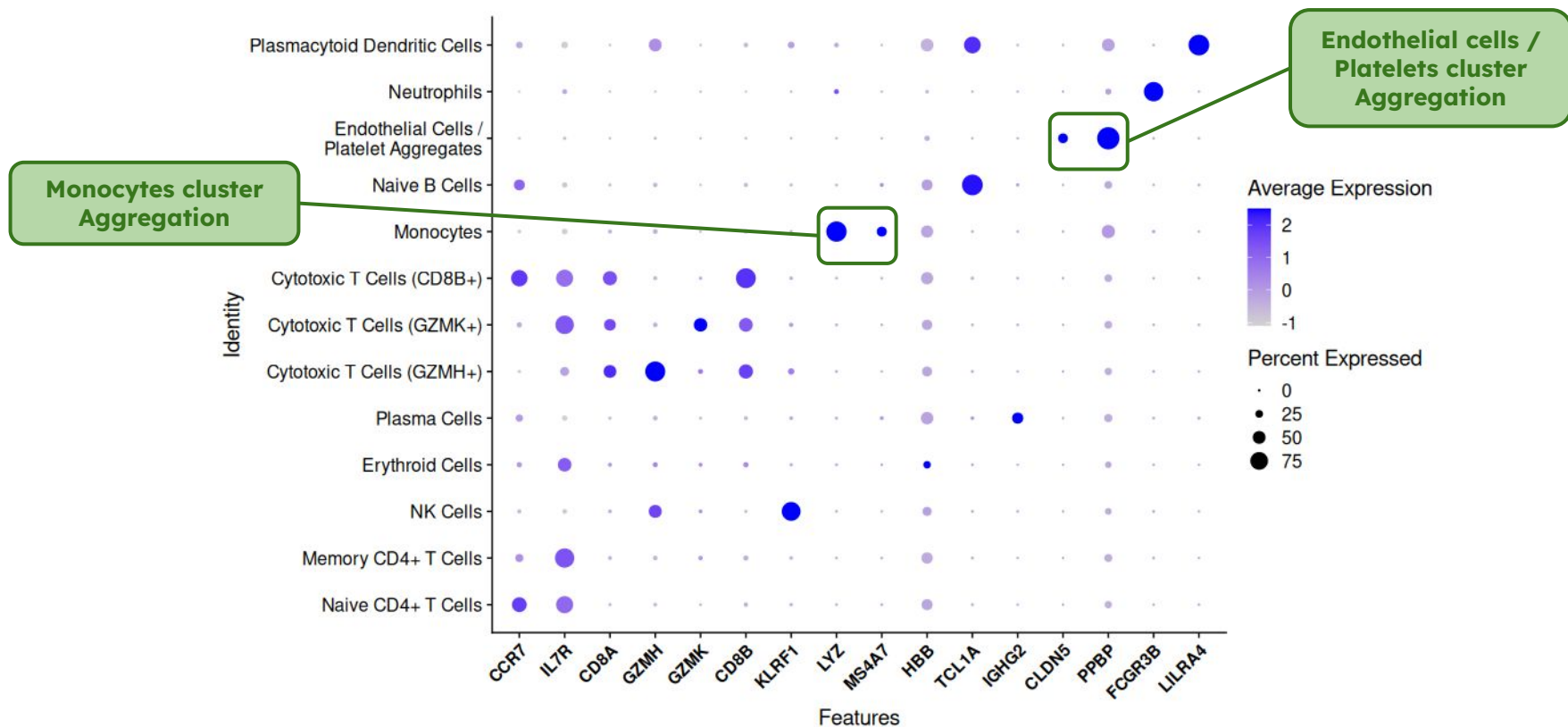


Marker Expression - Feature Plot



Gene	Assigned Cell type
IL7R	Naive/Memory CD4+ T Cells
CD8A	CD8+ T Cells
KLRF1	NK Cells
LYZ	CD14+ Monocytes
MS4A7	FCGR3A+ Monocytes
LILRA4	Plasmacytoid Dendritic Cells
PPBP	Platelets
TCL1A	Naive B Cells
IGHG2	Memory B Cells / Plasma Cells

Marker Expression - Dot Plot



Final results

